LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in Information Technology

Professional Installation Manual

Ubuntu Server Deployment and Initial Administration

Operational instructions for a junior system administrator deploying a repeatable Ubuntu Server lab environment.

Project Information	Details
Organization scenario	ABC Startup Solutions (hypothetical)
Course	ITEP 414 - System Administration and Maintenance
Target hostname	server01
VM baseline	4 GB RAM, 2 vCPUs, 40 GB virtual disk, NAT
Owner	[Your Full Name]
Version	1.0 - Student working draft

Prepared for ITEP 414 - System Administration and Maintenance

Working template: replace bracketed fields and add genuine lab evidence before submission.

Document Control and Scope

Item	Description
Purpose	Provide a repeatable process to install, configure, validate, and document Ubuntu Server.
Audience	Students, junior system administrators, and reviewers reproducing the lab.
In scope	Virtual machine setup, Ubuntu Server installation, hostname, DHCP networking, SSH, package updates, verification, troubleshooting, and evidence capture.
Out of scope	Production deployment approval, application installation, domain integration, firewall policy design, and high availability.

Use the guide with the actual Ubuntu Server ISO and virtualization platform available in your lab. Screen labels may differ slightly between installer releases. If your instructor supplies a specific ISO or procedure, follow that instruction and record the difference.

1. Prerequisites and Safety

- A supported host computer with virtualization enabled in firmware.
- Oracle VirtualBox or VMware Workstation installed and functioning.
- Ubuntu Server LTS ISO obtained from an official Ubuntu source.
- Enough free host resources for a 4 GB RAM, 2-vCPU, 40 GB virtual disk VM.
- A location to store screenshots and documentation.

Destructive operation warning: Guided - Use Entire Disk overwrites the selected target disk. In this lab, select only the newly created 40 GB virtual disk. Do not point the installer at a host disk containing personal or work data.

2. Reference Architecture

Component	Baseline	Reason
Virtual machine	Ubuntu-Server-Week03	Clear identification in the hypervisor.
CPU / memory	2 vCPUs / 4 GB RAM	Meets the project specification for the lab.
Disk	40 GB VDI/VMDK	Isolated virtual storage for the OS.
Network	NAT	Allows outbound connectivity while limiting unsolicited inbound access by default.
Hostname	server01	Stable, simple system identity.
Administrative user	Non-root user with sudo privileges	Reduces the need to log in directly as root.
Remote administration	OpenSSH Server	Enables encrypted remote terminal access when network access is configured.

3. Build the Virtual Machine

1. Create a new virtual machine named Ubuntu-Server-Week03.

2. Choose a 64-bit Ubuntu/Linux guest profile. Assign 4 GB RAM and 2 vCPUs.

3. Create a new 40 GB VDI or VMDK virtual disk. A dynamically allocated disk is suitable for a lab unless your instructor specifies otherwise.

4. Set the adapter to NAT unless instructed to use Bridged networking.
5. Attach the approved Ubuntu Server LTS ISO to the virtual optical drive.
6. Record the hypervisor version, firmware mode (BIOS/UEFI), ISO version, and VM settings for your documentation.

4. Install Ubuntu Server

4.1 Language and keyboard

Boot the VM from the ISO. Select English and the appropriate keyboard layout. Verify the keyboard mapping before entering the password.

4.2 Network and mirror

Accept DHCP unless the instructor requires static addressing. Record any address shown. Leave proxy and mirror values at their defaults unless your network requires a specific configuration.

4.3 Storage

Select Guided - Use Entire Disk, choose the new 40 GB virtual disk, review the proposed layout, and confirm only after verifying the target. Record the actual partition scheme, filesystem, and whether LVM is enabled. Do not assume the installer creates the same layout in every firmware or installer configuration.

4.4 Profile and SSH

Set the hostname to server01. Create a non-root administrative account with a strong, unique password. Select Install OpenSSH Server. Leave optional additional packages unselected unless required by the instructor.

4.5 Finish and reboot

Allow the installation to complete. Reboot when prompted and detach the ISO if necessary so the VM starts from its virtual disk. Log in with the account created during installation.

5. First-Boot Configuration

5.1 Confirm identity and network

```
hostname
ip addr
ip route
```

The hostname command should display server01. The ip addr command should show the network interfaces and assigned address; ip route should show the default route when one is configured. Record the active interface and the address assigned by DHCP.

5.2 Update package metadata and installed packages

```
sudo apt update
sudo apt upgrade -y
```

APT is the standard package-management interface for Ubuntu. Review output for failed repositories or dependency errors. Save the terminal output or screenshot. If the system indicates that a reboot is required, record that and reboot during an appropriate maintenance window.

5.3 Confirm SSH

```
systemctl status ssh
```

The expected state is active (running). If the service is missing and installation is permitted, run the following commands and record the change:

```
sudo apt install openssh-server
sudo systemctl enable --now ssh
systemctl status ssh
```

For remote access, confirm the target IP address and network path. NAT normally permits outbound traffic; inbound host-to-guest SSH may require an explicit port-forwarding rule. Do not expose SSH to the public internet for this classroom exercise.

6. Verification and Evidence Standard

ID	Test	Expected condition	Evidence
V-01	Console login	Created account authenticates successfully.	Screenshot of successful login.
V-02	hostname	Output equals server01.	Screenshot with command and output.
V-03	ip addr	Active interface has expected address.	Screenshot with relevant interface visible.
V-04	ping -c 4 google.com	Four replies are received, if ICMP is allowed by the network.	Screenshot of ping output.
V-05	sudo apt update; sudo apt upgrade -y	Commands finish without unresolved errors.	Screenshot or saved log.
V-06	systemctl status ssh	SSH service is active (running).	Screenshot of service status.

Use readable, genuine screenshots. Do not edit terminal output to make a failed test look successful. If a test fails, preserve the failure evidence, document the cause, apply a fix, and capture the re-test.

7. Troubleshooting Guide

Issue	Diagnostic approach	Possible next step
No address assigned	Inspect ip addr, adapter status, and DHCP settings.	Confirm the VM adapter is enabled and configured as required; restart networking only after checking the environment.
DNS resolution failure	Run resolvectl status and getent hosts google.com.	Check DNS configuration and upstream connectivity.
Ping fails but other sites work	Determine whether ICMP is blocked by the network.	Document the restriction and use an approved alternate connectivity test if your instructor allows it.
SSH service is inactive	Inspect systemctl status ssh and journalctl -u ssh.	Install OpenSSH Server if missing, then enable and start the service.
SSH is active but connection fails	Check IP, NAT/Bridged mode, firewall, and port-forwarding requirements.	Correct the network path; avoid opening unnecessary ports.
APT reports repository errors	Review the full error, system clock, DNS, and configured sources.	Correct the root cause before retrying package updates.
VM starts the installer again	Check attached ISO and boot order.	Eject the ISO or place the virtual disk first in boot order.

8. Operational Best Practices

- Keep the system updated and schedule reboots when updates require them.

- Use a non-root administrative account and sudo for privileged tasks.
- Use strong unique credentials; prefer SSH key-based authentication for environments that require remote access.
- Limit network exposure. In a production environment, define firewall rules, patching policy, logging, backups, and access reviews before offering services.
- Record the actual firmware mode, partition layout, IP address, service state, and software version.
- Do not commit passwords, private keys, tokens, or sensitive host details to a public repository.
- Capture evidence as soon as each verification task is complete.

9. Windows Server Evaluation Summary

For the bring-home activity, create a separate VM and install the official Windows Server 2025 Evaluation ISO unless the instructor specifies another release. A practical lab baseline is 2 vCPUs, 4 GB RAM, a 60 GB virtual disk, and NAT networking; check the current Microsoft system requirements before installation.

1. Create a VM named Windows-Server-Week03 and attach the evaluation ISO.

2. Choose the permitted evaluation edition. Select Desktop Experience if a graphical interface is required for the assignment.

3. Complete installation, assign a computer name, and create a strong Administrator password.

4. Log in successfully, run Windows Update, and capture screenshots of the installation and login.

5. Record the edition, version, hostname, VM settings, update status, and any issues encountered.

Microsoft's Evaluation Center lists a 180-day evaluation period for Windows Server 2025. Follow the current licensing and activation guidance displayed by Microsoft; do not use the evaluation as a permanent production license.

10. Evidence Checklist

Evidence item	Filename suggestion	Complete after capture
VM settings	01_vm_settings.png	[ ]
Ubuntu installer / storage summary	02_ubuntu_storage.png	[ ]
Console login	03_login.png	[ ]
Hostname	04_hostname.png	[ ]
IP address	05_ip_addr.png	[ ]
Connectivity test	06_ping_google.png	[ ]
System update	07_apt_update_upgrade.png	[ ]
SSH service	08_ssh_status.png	[ ]
Windows Server setup	09_windows_server_install.png	[ ]
Windows Server login	10_windows_server_login.png	[ ]

11. References

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